

CHAPTER 5

HYBRID DEEP LEARNING MODELS FOR ACCURATE AND COMPUTATIONALLY EFFICIENT WATER QUALITY PREDICTION IN AQUACULTURE

This chapter advances the research by developing hybrid CNN-LSTM/GRU models for WQP. This chapter introduces the implementation of these hybrid models and evaluates their performance on the ADAK and MAC WQ datasets. The results highlight the potential of these hybrid models in enhancing prediction accuracy and computational efficiency. CNN can effectively fetch the aquaculture WQ characteristics, whereas GRU and LSTM can learn long-term dependencies in the time series data. In Section 5.1 of this chapter, a comprehensive introduction is provided, highlighting the significant contributions of this work. Section 5.2 explains the architecture of the proposed CNN-LSTM/GRU models. A critical aspect of building an accurate DL WQP model is the identification of the suitable h_p . Section 5.3 presents an extensive study analysing the impact of various h_p variations on the performance of the proposed DL-RNN models. Different experiments are conducted using the two different WQ datasets and present an extensive study on the impact of h_p on the performance of the proposed hybrid DL models. Furthermore, the performance of hybrid CNN-LSTM and CNN-GRU models are compared with different baseline LSTM, GRU, and CNN DL models. Two different WQ datasets are utilized for the experimental evaluations to facilitate a comprehensive comparison, considering existing models such as LSTM, GRU, and CNN. Section 5.4 delves deeper into the performance analysis of all the DL-RNN models for a selected set of h_p , ultimately presenting the final results and comparisons. The results show that the hybrid CNN-LSTM outperformed all other models in terms of prediction accuracy and computation time. The chapter is summarized in Section 5.5.

5.1 INTRODUCTION

WQP plays an essential role in WQ management for aquaculture to make aquaculture production profitable and sustainable. It can be viewed as a sequential regression problem, aligning it with the structure of a typical time series challenge. Time series data has applications across fields like stock markets, healthcare, energy consumption, and weather forecasting [60–65]. Traditional methods for WQP, like ARMA, ARIMA,

SARIMA, and SVR, are unsuitable for aquaculture due to their linear assumptions and inefficiency in predicting non-linear water quality parameters [16–18].

The adaptability of DL models like LSTM, GRU, and CNN to the non-linear characteristics of aquaculture WQP is crucial [19]. These DL models surpass traditional RNNs and excel in various applications. Recent AI advancements have established DL as a proficient methodology, applied extensively in diverse fields [95, 96, 94]. DL enhances time series prediction precision and efficiency while accommodating non-linear water quality trends. Among DL models, LSTM stands out for its exceptional WQP performance [20]. However, LSTM and GRU models might struggle with extended memory retention in time series forecasting [97–99], hindering their capacity to capture prolonged dependencies.

Hybrid CNN-LSTM models function similarly to assist the LSTM model in capturing pertinent data. Utilizing a convolutional layer with maxpooling, the input sequence is compressed prior to utilization by the LSTM and GRU models. This integrated approach counters exponential decay; CNN with maxpooling inputs essential features, bolstering LSTM's forecasting accuracy for higher precision. These hybrid models prioritize accuracy, seeking to avoid overfitting or underfitting while demonstrating robust generalization. Moreover, their training demands reduced time. The convolutional layer, coupled with maxpooling, enhances the capacity of LSTM and GRU to learn and retain pivotal features within non-linear water quality parameters, mitigating exponential decay through convolution and maxpooling [92, 114]. Notably, CNNs inherently operate faster, processing computations in parallel, providing an advantage over RNNs requiring sequential processing. Quasi-Recurrent Neural Networks [115] emulate RNN behaviour using CNN components, effectively boosting speed. Furthermore, CNNs focus on crucial features, leading to complexity reduction. By incorporating convolution layers with Maxpooling, the tensor size is diminished, which accelerates training for hybrid models. These hybrid CNN-LSTM models find applications in various research endeavors involving time series prediction. DL models, widely adopted across different time series domains, encompass finance (stock, cryptocurrency, precious metal prediction) [116, 117]. For instance, the CNN-LSTM hybrid, utilized for gold price prediction [118], exhibits efficacy relative to other DL models. The employment of Bi-LSTM with CNN for air quality prediction [120] achieves high accuracy in air quality estimation. Hybrid DL models are prevalent in air quality (PM_{2.5}) prediction, leveraging CNN and LSTM [121, 122].

In this work, a hybrid DL models, CNN with the LSTM and GRU for aquaculture WQP is proposed. CNN can effectively fetch the aquaculture WQ characteristics,

whereas GRU and LSTM can learn long-term dependencies in the time series data. Different experiments are conducted using the two different WQ datasets and present an extensive study on the impact of hyperparameters on the performance of the proposed hybrid DL models. Furthermore, the performance of hybrid CNN-LSTM and CNN-GRU models are compared with different baseline LSTM, GRU and CNN DL models.

The significant contributions of the research work presented in this chapter are as follows:

1. This work pioneers the application of hybrid CNN-LSTM and CNN-GRU models in the realm of WQP. These hybrid models fuse the feature extraction prowess of CNNs with the long-term dependency learning capabilities of LSTM/GRU networks, yielding a novel approach to WQP.
2. Developed hybrid models, meticulously designed to predict critical WQ parameters. These parameters, including salinity, pH, DO, and temperature, are vital indicators in aquaculture.
3. Extensive experimental evaluations are conducted to investigate the impact of h_p on the performance of the proposed hybrid CNN-LSTM, CNN-GRU, and all DL-RNN models, using two different aquaculture WQ datasets. Determining the right set of h_p plays a significant role in building an accurate DL-RNN WQP model.
4. The hybrid models' performance is rigorously evaluated using real-world sensor data from two different WQ datasets. The results underline the models' potential to significantly improve the accuracy of WQP, thereby enhancing the reliability of monitoring systems in aquaculture settings.
5. The performance of hybrid CNN-LSTM and CNN-GRU models are compared with baseline DL models LSTM, GRU, and CNN. Results show that the hybrid models significantly improve accuracy compared to the baseline models for aquaculture WQP. Also, the hybrid models outperform all the DL models in terms of computation time.

5.2 PROPOSED HYBRID CNN-LSTM/GRU DL-RNN MODELS

The Hybrid CNN-LSTM and CNN-GRU DL-RNN structures are shown in Fig. 5.1 and Fig. 5.2. The two models are the hybrid of CNN with LSTM and CNN with GRU. The first part of the model is CNN, to which the data is fed, and it extracts the

features. There is a dropout layer after the convolution layer (Conv1D) and pooling layer (MaxPooling1D). The second part has the LSTM or GRU, followed by a dense layer to give the output.

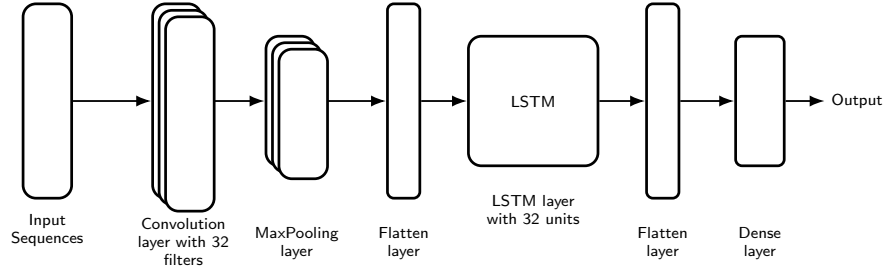


Figure 5.1: Proposed Hybrid CNN-LSTM DL-RNN model structure

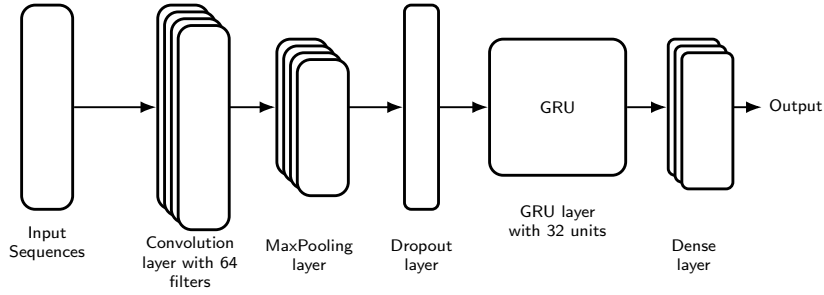


Figure 5.2: Proposed Hybrid CNN-GRU DL-RNN model structure

In the CNN-LSTM model, Conv1D with 32 filters is used, a kernel size of 3 and ReLU is used as the activation function. A pooling layer MaxPooling1D follows this with a pooling size of 2. The CNN layer extracts all the features and then feeds them to the LSTM layer. The output of the CNN layer is fed to the LSTM layer after pooling through the flatten layer. Then, the LSTM layer outputs its output to the dense layer through the flatten layer, and finally, the prediction is output at the dense layer.

The CNN-GRU models use Conv1D with 64 filters, and a kernel size of 5 and ReLU is used as the activation function. A pooling layer MaxPooling1D follows this with a pooling size of 4. The CNN layer extracts all the features and then feeds them to the GRU layer. The output of the CNN layer is fed to the GRU layer after pooling and 0.2 dropout. After that, the GRU layer outputs its output to the dense layer, and finally, the prediction is output at the dense layer. These models use a learning rate of 0.0008 and Adam for optimisation [89]. Here, MSE is used as the loss function. The algorithm is based on CNN, LSTM, and GRU DL-RNN implemented using Python, Keras, and TensorFlow.

5.2.1 Model implementation

The WQ dataset forms the basis for training the hybrid CNN-LSTM/GRU DL-RNN models for WQP. The dataset is initially split into two sections: 80% for training and 20% for testing. The training dataset aids in model development, while the testing dataset is essential for validating and comparing the performance of the developed DL-RNN models. To ensure effective training, a key preprocessing step involves scaling all input and output variables within the range of 0 to 1, achieved through normalization and minimum-maximum scaling techniques. This scaling is facilitated using the `MinMaxScaler` from the `scikit-learn` preprocessing library in Python [146]. In addition, for model training, the Adam optimizer is strategically chosen due to its adaptive learning rate mechanism and adeptness in handling sparse gradients [147]. This choice of optimizer contributes to the efficiency of training the DL-RNN models. The training and testing datasets then undergo a transformation into supervised learning frames through various array manipulation techniques. Subsequently, the time series sequences are translated into input-output equations using the sliding window procedure. This transformed data is used for effectively training the DL models for WQP.

5.3 EXPERIMENTS: HYPERPARAMETER TUNING

This research is to analyse the performance of hybrid models CNN-LSTM and CNN-GRU. Two WQ data sets are used to conduct experiments, analyse and evaluate the performance of the proposed models. Furthermore, by comparing proposed models with other DL models, the prediction performance and effectiveness of the proposed models are validated. All experiments are conducted on Microsoft Azure Virtual Machines with specifications: Inter(R) Xeon (R) 8272CL CPU @2.60GHz, 32 GB RAM. The open source software used are Python and Keras based on Tensorflow.

The hybrid models are compared with CNN, LSTM, and GRU DL models. In the data set, 80% of the data is used to train the model, and the remaining 20% to test the model's prediction accuracy. The most critical part of building a DL model is tuning and optimization of the h_p . For the model to be most effective and efficient, tuning as many h_p as possible is necessary. In this work, four h_p are selected for tuning: learning rate, epochs, window size, and batch size. In each of the models, the number of layers and neurons in each layer are fixed. A dropout of 0.2 is applied after the primary layers to avoid overfitting, and *ReLU* is applied as the activation function. Additionally, the optimizer adopted in all the DL models in this work is Adam.

This work aims to improve the accuracy of aquaculture WQP and reduce computation time. Four sets of experiments are conducted involving varying of each h_p . Learning rate, epoch, window size, and batch size are plotted against RMSE. Additionally,

each hyperparameter is plotted against computation time. The experiments encompass both datasets (ADAK and MAC). This aids in analysing DL model performance on medium-sized and large datasets. The ADAK WQ dataset, despite spanning three years, only comprises 1096 data points, recorded once daily. In contrast, the MAC WQ dataset has 23200 data points, representing 80 days of recordings, taken every 5 minutes.

5.3.1 Learning rate (L_r)

The impact of learning rate $\{0.01, 0.001, 0.0009, 0.0008, 0.0007, 0.0001\}$ on the performance for each of the DL models is analysed. In Fig. 5.3, RMSE vs learning rate performance is compared between proposed hybrid DL models and baseline models using the ADAK WQ dataset. The proposed hybrid DL models perform consistently better than baseline DL models. However, the performance of baseline DL models is not consistent for various learning rates and different WQ parameters. Hence using the baseline DL models for WQP is not practical.

To further analyse the performance of these models, the experiment is repeated with the MAC WQ dataset. In Fig. 5.4, RMSE vs learning rate performance of proposed hybrid DL models is compared with baseline models using the MAC WQ dataset. LSTM, GRU and CNN-GRU show moderate performance at some learning rates. However, the performance of CNN does not come close to the performance of hybrid models at any point. Here, it can be observed from Fig. 5.4 that the hybrid CNN-LSTM model has the best and consistent performance compared to other models.

5.3.2 Epoch (E_p)

The outcome of choosing different epochs $\{10, 50, 100, 150, 200, 300, 400, 500\}$ for each DL model is studied. Figure 5.5 shows the RMSE vs epoch for different WQ parameters using ADAK WQ dataset for the proposed hybrid DL models and LSTM, GRU as well as CNN DL models. Here the proposed hybrid DL models CNN-LSTM and CNN-GRU maintain better performance than baseline DL models LSTM, GRU and CNN. Hybrid models achieve good performance within 50 to 100 epochs, and the generalization is attained within 100 epochs. Prediction accuracy performance is maintained for all four WQ parameters for epochs 10 to 500. Also, the prediction performance of our hybrid DL models is higher compared to the baseline DL models. As well as baseline DL models are taking at least 400 epochs to achieve generalization and is under-fitting training data for epochs less than 400. The baseline DL models are close to the performance of the proposed models at 500 epochs. The baseline models require more computation resources to achieve similar results. The hybrid models start

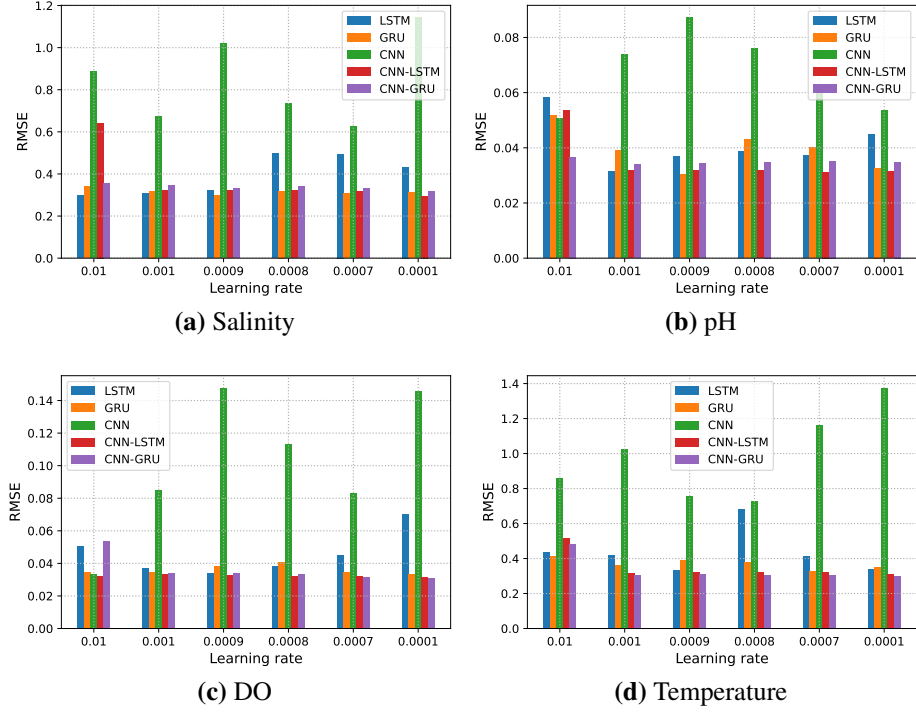


Figure 5.3: Comparison of prediction accuracy (RMSE) of DL-RNN models for various L_r and fixed h_p ($E_p = 100$, $W_z = 30$, $B_z = 32$) on the ADAK water quality dataset

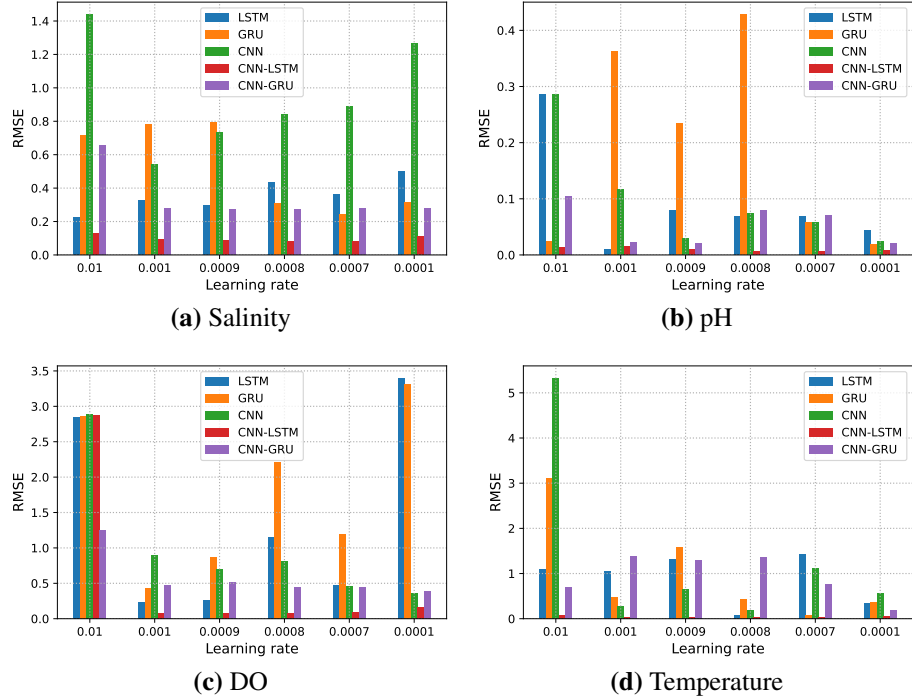


Figure 5.4: Comparison of prediction accuracy (RMSE) of DL-RNN models for various L_r and fixed h_p ($E_p = 50$, $W_z = 30$, and $B_z = 64$) on the MAC water quality dataset

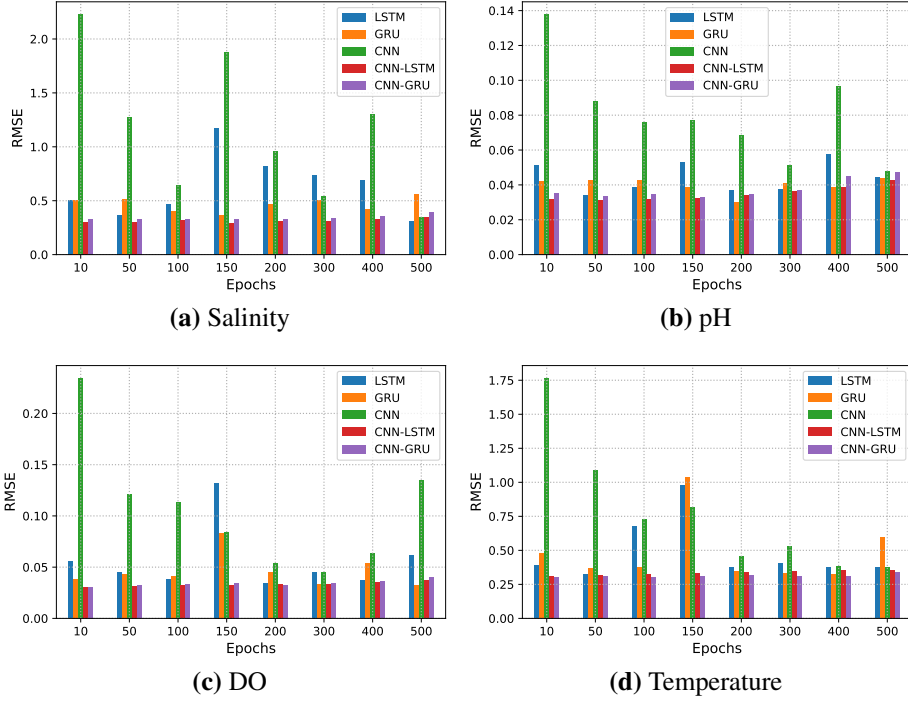


Figure 5.5: Comparison of prediction accuracy (RMSE) of DL-RNN models for various E_p and fixed h_p ($L_r = 0.0008$, $W_z = 30$, $B_z = 32$, and O_p is Adam) on the ADAK water quality dataset

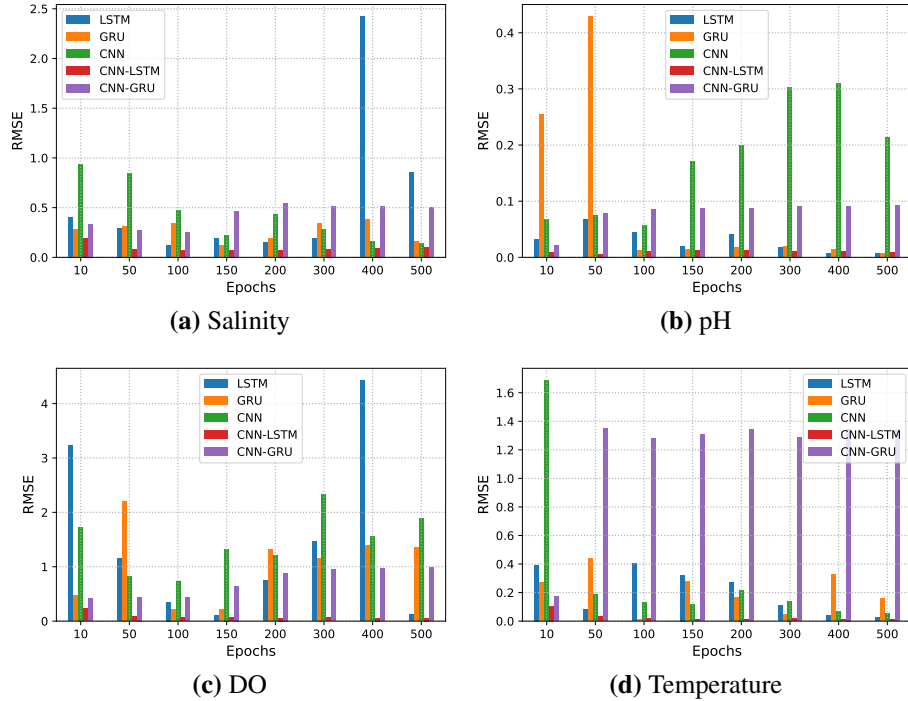


Figure 5.6: Comparison of prediction accuracy (RMSE) of DL-RNN models for various E_p and fixed h_p ($L_r = 0.0008$, $W_z = 30$, $B_z = 64$, and O_p is Adam) on the MAC water quality dataset

to overfit a little after 150 epochs for the ADAK WQ dataset. Still, the accuracy is maintained, indicating the adaptability of the proposed DL model to different datasets.

To further evaluate the performance, the experiment is repeated with the MAC WQ dataset. Figure 5.6 shows the RMSE vs epoch for all four WQ parameters on the MAC dataset of the proposed hybrid DL models, LSTM, GRU and CNN. Here the proposed hybrid DL models CNN-LSTM maintain better performance than the other DL models for predictions of WQ parameters. In this dataset, our model attained the required accuracy and generalization in 50 epochs. Furthermore, the accuracy loss is minimal after increasing epochs, and the CNN-LSTM model is neither over-fitting nor under-fitting the training data as the epochs is increased from 10 to 500 epochs.

5.3.3 Window size (W_z)

The influence of choosing different window size $\{10, 20, 30, 40, 50, 60, 70, 80\}$ for each DL model is analysed. Figure 5.7 shows the RMSE vs window size for different WQ parameters on the ADAK WQ dataset with the proposed hybrid DL models and the baseline DL models. The proposed hybrid DL models perform better than baseline DL models. As the window size is increased from 10 to 80, the error remains constant for hybrid DL models. In this experiment, it can be seen that the baseline models are comparable in performance with hybrid DL models for some window sizes for each WQ parameter. However, no baseline models are performing consistently well for all four WQ parameters. For example, the performance of LSTM is slightly better than CNN-LSTM at window size 40 and 80 for pH. However, for the other three WQ parameters, the performance of LSTM is not good. In comparison, proposed hybrid models consistently show good performance for the four WQ parameters and the window sizes experimented with.

To further analyse the performance of these models, the experiment is repeated with the MAC WQ dataset. Figure 5.8 shows the RMSE vs window size for all four WQ parameters on the MAC dataset with the proposed hybrid DL models, LSTM, GRU as well as CNN. Furthermore, it can be observed from Fig. 5.8 that the hybrid CNN-LSTM model is having good performance consistently when compared to the other models.

5.3.4 Batch size (B_z)

In this experiment, the performance of each DL model is analysed on different batch sizes $\{16, 32, 64, 128, 256, 512\}$. Figure 5.9 shows the RMSE vs batch size for different WQ parameters on the ADAK WQ dataset for the proposed hybrid DL models with the baseline DL models. Compared to the proposed hybrid DL models, it can be

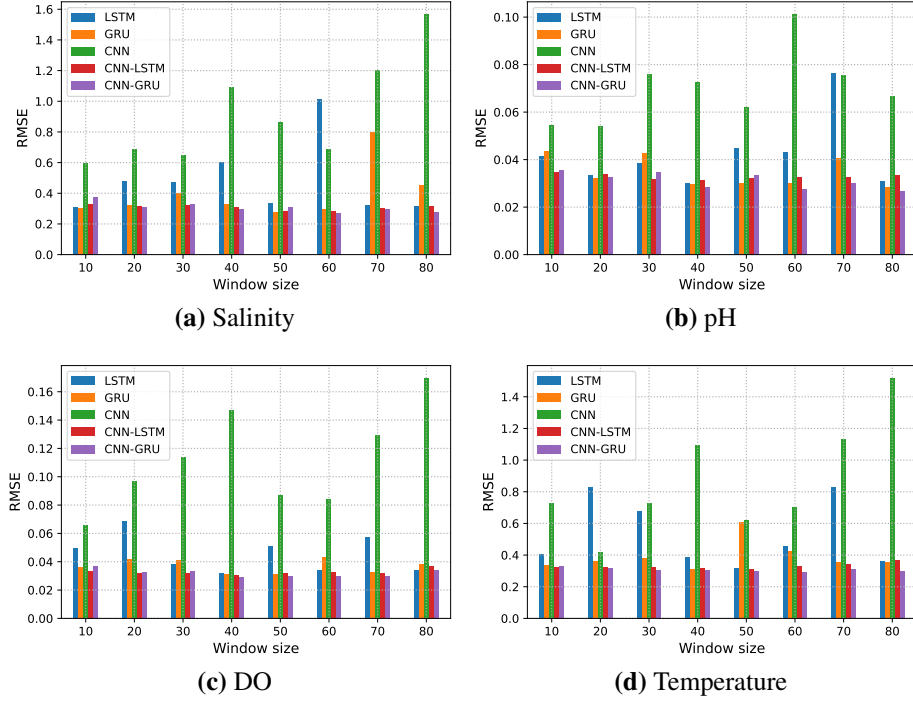


Figure 5.7: Comparison of prediction accuracy (RMSE) of DL-RNN models for various W_z and fixed h_p ($L_r = 0.0008$, $E_p = 100$, $B_z = 32$ and O_p is Adam) on the ADAK water quality dataset

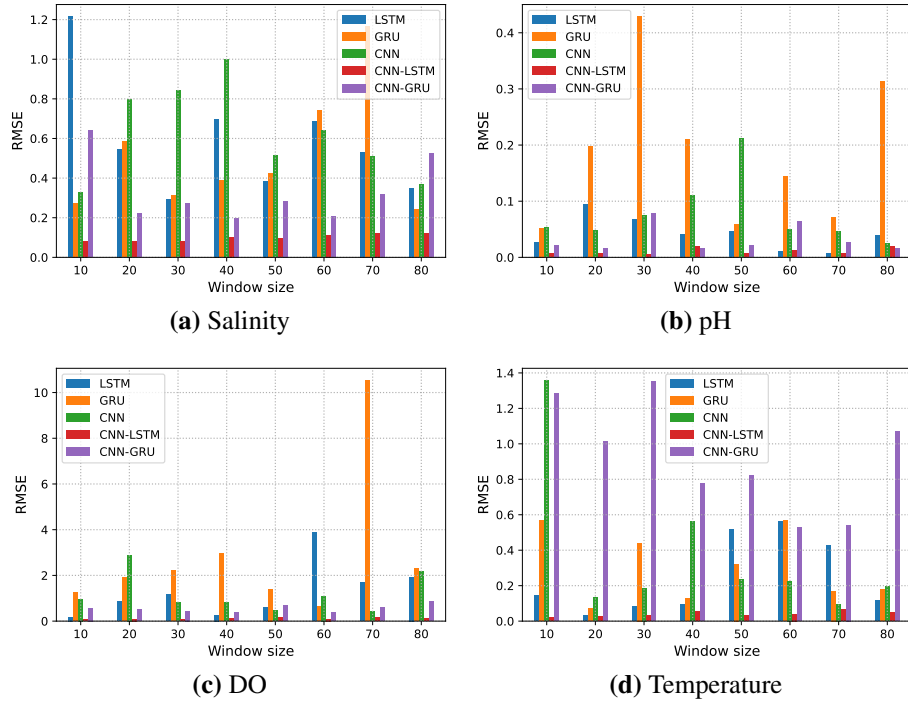


Figure 5.8: Comparison of prediction accuracy (RMSE) of DL-RNN models for various W_z and fixed h_p ($L_r = 0.0008$, $E_p = 50$, $B_z = 64$ and O_p is Adam) on the MAC water quality dataset

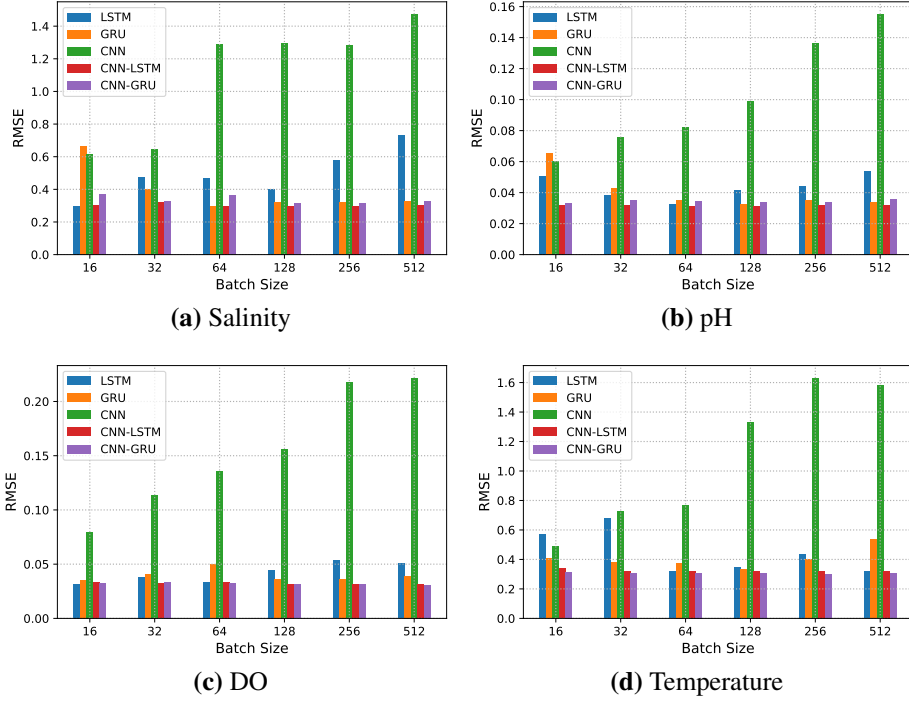


Figure 5.9: Comparison of prediction accuracy (RMSE) of DL-RNN models for various B_z and fixed h_p ($L_r = 0.0008$, $E_p = 100$, $W_z = 30$, and O_p is Adam) on the ADAK water quality dataset

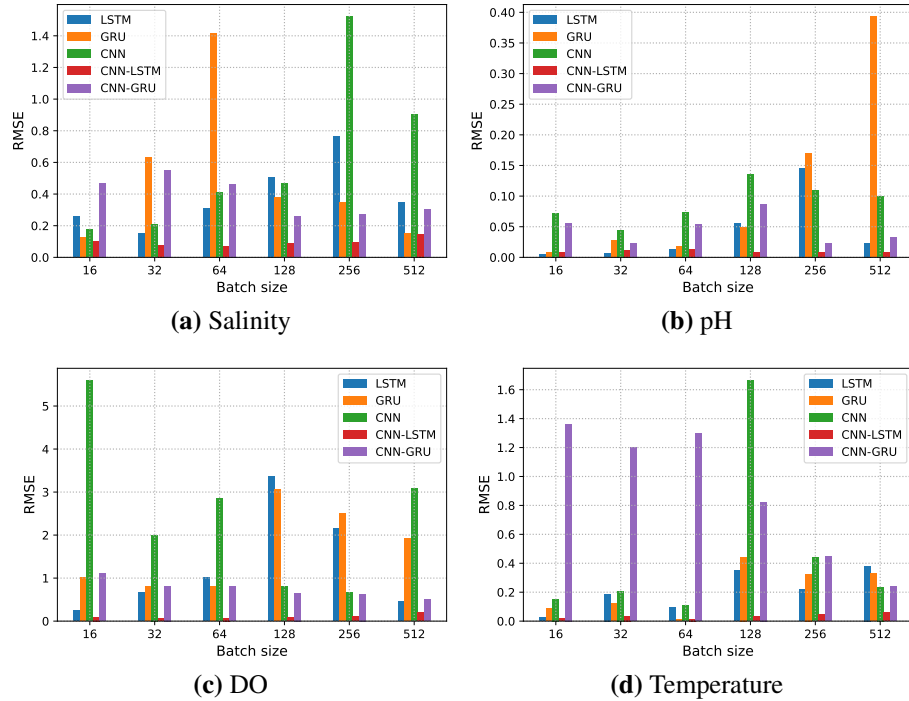


Figure 5.10: Comparison of prediction accuracy (RMSE) of DL-RNN models for various B_z and fixed h_p ($L_r = 0.0008$, $E_p = 100$, $W_z = 30$, and O_p is Adam) on the MAC water quality dataset

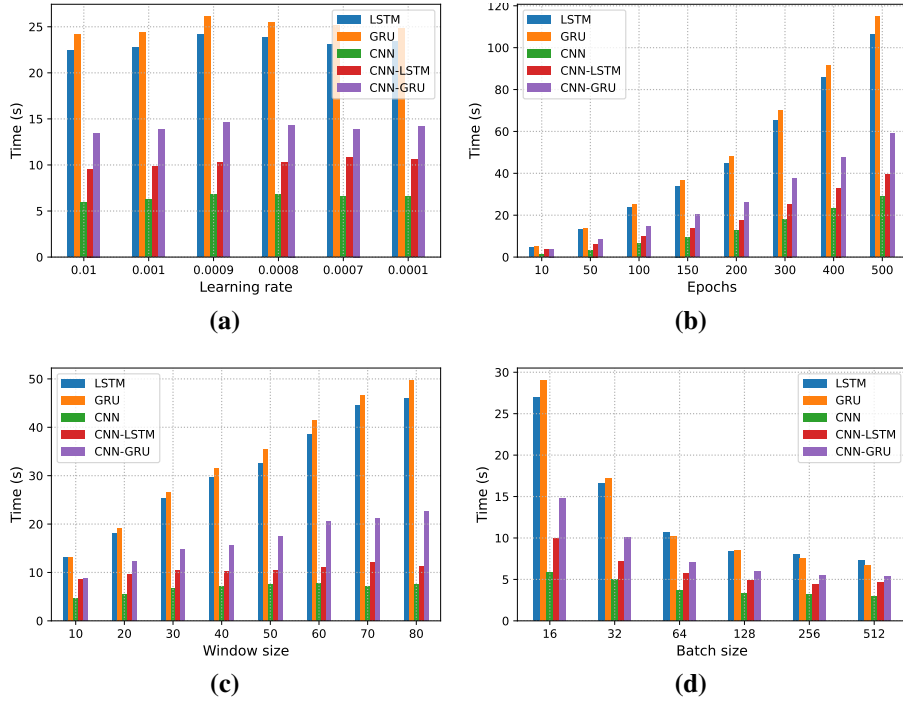


Figure 5.11: Comparison of computation time for prediction using all DL-RNN models. (a) $L_r = \{0.01, 0.001, 0.0009, 0.0008, 0.0007, 0.0001\}$, (b) $E_p = \{10, 100, 150, 200, 300, 400, 500\}$, (d) $W_z = \{10, 20, 30, 40, 50, 60, 70, 80\}$, and (e) $B_z = \{16, 32, 64, 128, 256, 512\}$ on ADAK water quality dataset

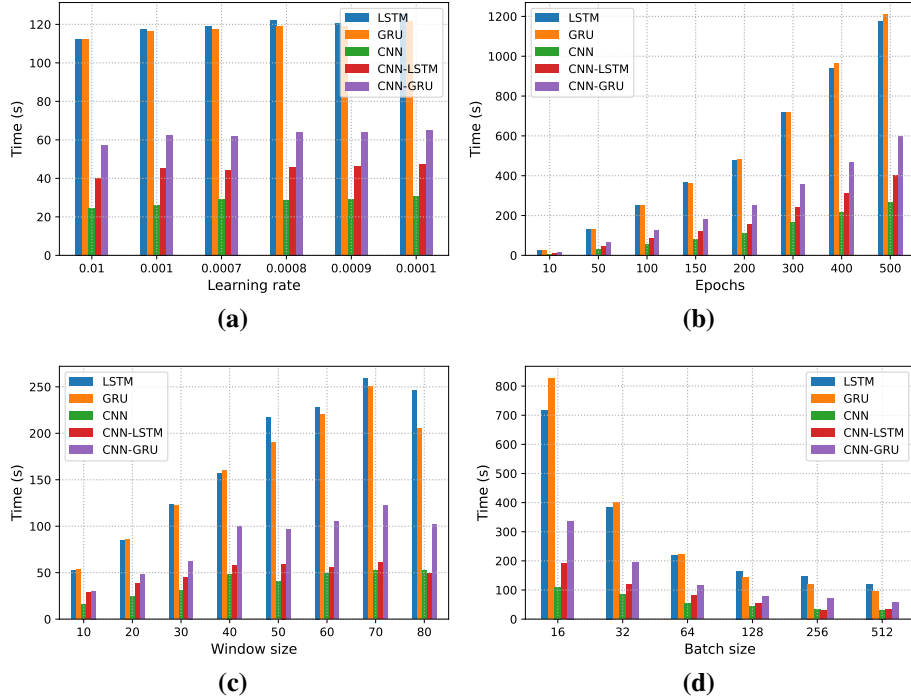


Figure 5.12: Comparison of computation time for prediction using all DL-RNN models. (a) $L_r = \{0.01, 0.001, 0.0009, 0.0008, 0.0007, 0.0001\}$, (b) $E_p = \{10, 100, 150, 200, 300, 400, 500\}$, (d) $W_z = \{10, 20, 30, 40, 50, 60, 70, 80\}$, and (e) $B_z = \{16, 32, 64, 128, 256, 512\}$ on MAC water quality dataset

observed that the performance of the baseline DL models is inconsistent and inferior. Furthermore, it can be seen that the performance of CNN is reducing with the increase of the batch size.

Figure 5.10 shows the RMSE vs batch size for different WQ parameters using the MAC WQ dataset with the proposed hybrid DL models and the baseline DL models. As this dataset has more data points, it can be seen from the plots the performance difference of each model. From the results, it can be observed that the proposed hybrid CNN-LSTM models have better performance compared to other baseline models and the hybrid CNN-GRU model. Here also, the baseline DL models are inconsistent and underperform compared to the proposed hybrid DL models.

5.3.5 Computation time

In the above four experiments, the computation time was also simultaneously measured and stored. Figure 5.11 and Fig. 5.12 plot the computation time for each hyperparameter for the ADAK WQ dataset and MAC WQ dataset, respectively. The computation time required by each DL model is different. Here some common trends from the plots can be observed. One among them is that the computation time increases with an increase in epochs and window size. But computation time is reduced with an increase in batch size. Significant changes in computation time based on different learning rates are not observable due to the constrained range of learning rates applied consistently across all models for both datasets.

In summary, for WQP, the proposed CNN-LSTM model maintains the superior performance in various experimental conditions in terms of accuracy and computation time, indicating the adaptability of the proposed CNN-LSTM DL model.

5.4 RESULTS AND DISCUSSIONS

The aim is to assess how effectively hybrid DL models, CNN-LSTM and CNN-GRU, perform compared to LSTM, GRU, CNN DL models to predict aquaculture WQ parameters. In this section, the comparison involves assessing the performance of proposed hybrid DL models against the baseline DL models. The evaluation employs a fixed set of h_p across the two datasets.

The h_p is selected by studying the performance of the DL models for various hyperparameters (epochs, window size, learning rate and batch size) in terms of prediction accuracy and computation time. The performance of each model is studied through various experiments by varying the hyperparameters. The optimal set of hyperparameters in terms of prediction accuracy and computation time is selected to run the final

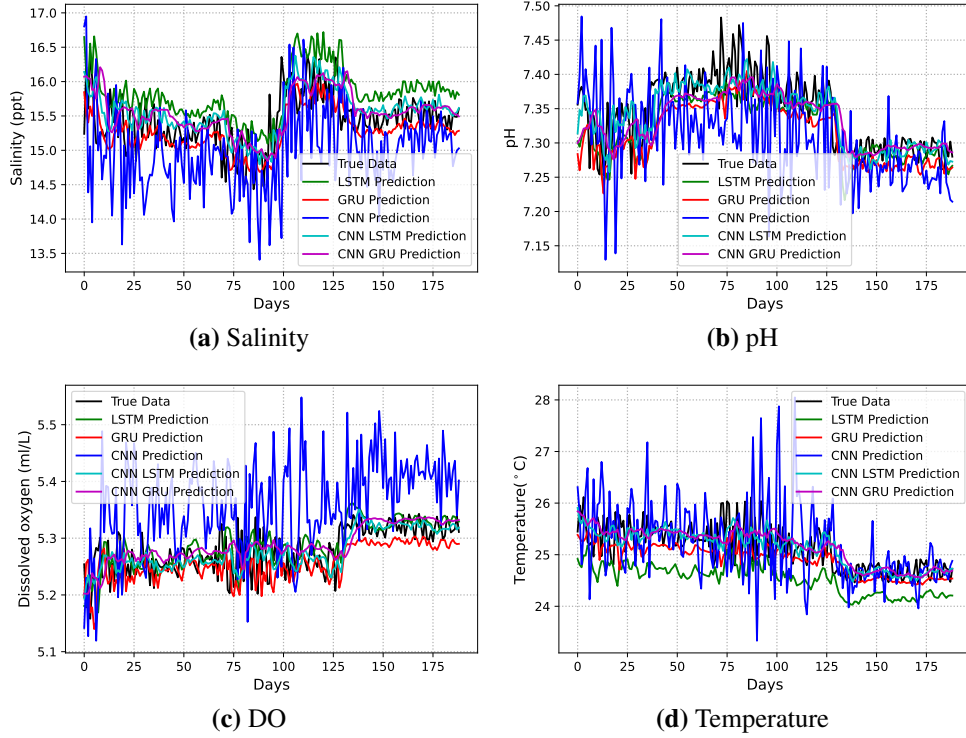


Figure 5.13: Comparison of predicted values using LSTM, GRU, CNN, CNN-LSTM, and CNN-GRU predictions and ADAK water quality observed data. (a) Salinity, (b) pH, (c) DO, (d) Temperature

experiment for comparing the performance of all the DL models.

5.4.1 ADAK water quality dataset

Figure 5.13 compares the predicted values with the true data on the ADAK WQ dataset using different DL models. All these models are trained using 80% of the ADAK WQ dataset and tested with 20% of the data. The predicted values are compared with the true data. Here a set of h_p (epochs = 100, learning rate = 0.0008, batch size = 32, window size = 30 and Adam optimizer) is fixed. The models are evaluated using performance metrics MAE, MSE, RMSE, and MAPE in the training and testing periods. The computation time required for each of the models is also calculated.

For the ADAK WQ dataset, the prediction performance of all models is shown in Table 5.1. From the results its clear that the proposed CNN-LSTM model (MAE = 0.2509 ppt, MSE = 0.1035 ppt, RMSE = 0.3217 ppt and MAPE = 0.0164 ppt) outperforms all baseline DL models in the training and testing period. For the salinity prediction, the LSTM model (MAE = 0.3942 ppt, MSE = 0.2230 ppt, RMSE = 0.4722 ppt and MAPE = 0.0257 ppt) outperforms the CNN model (MAE = 0.4800 ppt, MSE = 0.4180 ppt, RMSE = 0.6466 ppt and MAPE = 0.0312 ppt). When the performance of WQP is compared with other works [148], a similar trend can be observed where

Table 5.1: Prediction accuracy of LSTM, GRU, CNN, CNN-LSTM and CNN-GRU for ADAK water quality dataset

Water Quality Parameter	Model	Training Period					Testing Period				
		MAE	MSE	RMSE	MAPE	Time (s)	MAE	MSE	RMSE	MAPE	Time (s)
Salinity (ppt)	LSTM	0.2741	0.1287	0.3588	0.0184	23.47	0.3942	0.2230	0.4722	0.0257	0.05
	GRU	0.3550	0.1843	0.4293	0.0237	25.02	0.3270	0.1614	0.4017	0.0212	0.05
	CNN	0.4180	0.2911	0.5396	0.0282	05.55	0.4800	0.4180	0.6466	0.0312	0.04
	CNN-LSTM	0.2274	0.0917	0.3028	0.0153	10.27	0.2509	0.1035	0.3217	0.0164	0.04
	CNN-GRU	0.2434	0.1066	0.3264	0.0164	14.00	0.2475	0.1100	0.3317	0.0161	0.04
pH	LSTM	0.0321	0.0017	0.0409	0.0044	22.72	0.0300	0.0015	0.0386	0.0041	0.05
	GRU	0.0341	0.0018	0.0427	0.0046	25.32	0.0353	0.0018	0.0429	0.0048	0.05
	CNN	0.0498	0.0037	0.0608	0.0068	05.49	0.0600	0.0058	0.0759	0.0082	0.04
	CNN-LSTM	0.0220	0.0009	0.0293	0.0030	10.37	0.0237	0.0010	0.0317	0.0032	0.04
	CNN-GRU	0.0259	0.0012	0.0345	0.0035	14.93	0.0269	0.0012	0.0349	0.0037	0.04
DO (ml/L)	LSTM	0.0290	0.0014	0.0374	0.0055	24.64	0.0295	0.0015	0.0384	0.0056	0.05
	GRU	0.0331	0.0017	0.0414	0.0062	25.49	0.0319	0.0017	0.0411	0.0060	0.05
	CNN	0.1072	0.0151	0.1229	0.0201	06.13	0.0974	0.0129	0.1135	0.0185	0.04
	CNN-LSTM	0.0226	0.0009	0.0292	0.0042	10.68	0.0250	0.0010	0.0321	0.0047	0.04
	CNN-GRU	0.0258	0.0012	0.0346	0.0049	14.21	0.0262	0.0011	0.0334	0.0050	0.04
Temperature(°C)	LSTM	0.5552	0.4080	0.6388	0.0222	24.40	0.6071	0.4633	0.6807	0.0240	0.05
	GRU	0.2852	0.1357	0.3684	0.0114	24.88	0.2951	0.1444	0.3800	0.0116	0.05
	CNN	0.5346	0.5422	0.7363	0.0217	05.51	0.5229	0.5295	0.7276	0.0207	0.04
	CNN-LSTM	0.2252	0.0846	0.2908	0.0091	10.60	0.2471	0.1036	0.3219	0.0098	0.04
	CNN-GRU	0.2560	0.1162	0.3409	0.0103	14.08	0.2410	0.0938	0.3063	0.0096	0.04

h_p : $L_r = 0.0008$, $E_p = 100$, $W_z = 30$, $B_z = 32$, and Adam optimizer.

LSTM have outperformed CNN in prediction accuracy. However, it is noticeable that the computation time required for training CNN is just 5.55s which is only 23.64% the computation time required by LSTM, which requires 23.72s. The LSTM model has the ability to learn from long-term dependencies, and CNN can learn features from a large dataset. However, the hybrid CNN-LSTM requires a computation time of 10.27s only for training the model, which is 43.75% of the computation time of LSTM saving 56.24% in computation time and outperforming all other models in terms of prediction accuracy.

For pH also, the CNN-LSTM model (MAE = 0.0237, MSE = 0.0010, RMSE = 0.0317 and MAPE = 0.0032) outperforms all other baseline DL models in terms of prediction accuracy. The same performance is repeated for DO (MAE = 0.0250 ml/L, MSE = 0.0010 ml/L, RMSE = 0.0321 ml/L, and MAPE = 0.0047 ml/L) and temperature (MAE = 0.2471 °C, MSE = 0.1036 °C, RMSE = 0.3219 °C, and MAPE = 0.0098 °C) prediction as well. Also, the computation time required for CNN-LSTM is around 45% of the computation time needed by LSTM and GRU models. CNN-LSTM outperforms CNN with prediction accuracy even though the CNN model requires lesser computational time when compared to CNN-LSTM. Hence, it can be concluded that CNN-LSTM is the DL model providing better performance for ADAK WQ data.

5.4.2 MAC water quality dataset

Figure 5.14 compares the prediction of all DL models with the true data on the MAC WQ dataset. All these models are trained using 80% of the ADAK WQ dataset and

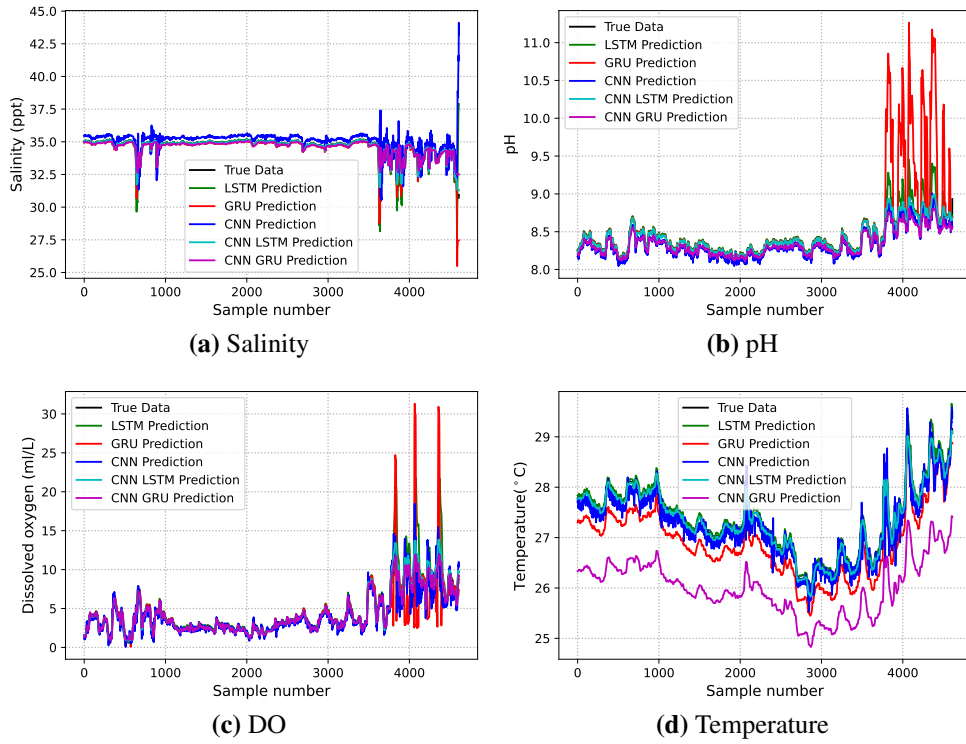


Figure 5.14: Comparison of predicted values using LSTM, GRU, CNN, CNN-LSTM, and CNN-GRU predictions and MAC water quality observed data.(a) Salinity, (b) pH, (c) DO, (d) Temperature

tested on the 20% values. The predictions are compared with the true data. Here a set of h_p (epochs = 50, learning rate = 0.0008, batch size = 64, window size = 30 and Adam optimizer) is fixed. The models are evaluated using MAE, MSE, RMSE, and MAPE in the training and testing periods. The computation time required for each of the models is also calculated.

For the MAC WQ dataset, the prediction performance of all models is shown in Table 5.2. From the table, it is clear that the proposed CNN-LSTM model (MAE = 0.0381 ppt, MSE = 0.0068 ppt, RMSE = 0.0827 ppt and MAPE = 0.0011 ppt) outperforms all baseline DL models in the training and testing period. For the salinity prediction, the LSTM model (MAE = 0.0535 ppt, MSE = 0.0860 ppt, RMSE = 0.2933 ppt and MAPE = 0.016 ppt) outperforms the CNN model (MAE = 0.4783 ppt, MSE = 0.7089 ppt, RMSE = 0.8240 ppt and MAPE = 0.0141 ppt). But it is noticeable that the computation time required for training CNN is just 21.88s which is only 19.57% the computation time required by LSTM, which requires 111.75s. The hybrid CNN-LSTM requires a computation time of 36.83s only for training the model, which is 32.95% of LSTM saving 67.04% in computation time and outperforming all other models in terms of prediction accuracy.

For pH also, the CNN-LSTM model (MAE = 0.0042, MSE = 0.00004, RMSE =

Table 5.2: Prediction accuracy of LSTM, GRU, CNN, CNN-LSTM and CNN-GRU for MAC water quality dataset

Water Quality Parameter	Model	Training Period					Testing Period				
		MAE	MSE	RMSE	MAPE	Time (s)	MAE	MSE	RMSE	MAPE	Time (s)
Salinity (ppt)	LSTM	0.0173	0.0010	0.0320	0.0005	111.75	0.0535	0.0860	0.2933	0.0016	0.35
	GRU	0.0210	0.0007	0.0258	0.0006	114.13	0.0571	0.0975	0.3123	0.0018	0.31
	CNN	0.3612	0.1455	0.3815	0.0104	021.88	0.4783	0.7089	0.8420	0.0141	0.11
	CNN-LSTM	0.0241	0.0011	0.0338	0.0007	36.83	0.0381	0.0068	0.0827	0.0011	0.17
	CNN-GRU	0.0989	0.0159	0.1259	0.0029	055.74	0.1460	0.0736	0.2714	0.0043	0.19
pH	LSTM	0.0166	0.0003	0.0175	0.0020	116.86	0.0339	0.0047	0.0689	0.0039	0.35
	GRU	0.0041	0.0001	0.0101	0.0005	115.93	0.1358	0.1839	0.4288	0.0154	0.31
	CNN	0.0611	0.0042	0.0645	0.0074	022.56	0.0696	0.0055	0.0744	0.0083	0.11
	CNN-LSTM	0.0031	0.00002	0.0045	0.0004	039.07	0.0042	0.00004	0.0063	0.0005	0.17
	CNN-GRU	0.0429	0.0024	0.0492	0.0051	058.74	0.0673	0.0063	0.0792	0.0079	0.19
DO (ml/L)	LSTM	0.3148	0.1310	0.3620	0.0803	116.31	0.4767	1.3341	1.1550	0.0873	0.36
	GRU	0.2235	0.0762	0.2760	0.0569	119.47	0.6789	4.8710	2.2070	0.0785	0.32
	CNN	0.3300	0.1454	0.3813	0.0866	023.20	0.4492	0.6705	0.8188	0.0952	0.11
	CNN-LSTM	0.0467	0.0033	0.0572	0.0129	040.76	0.0566	0.0065	0.0804	0.0145	0.18
	CNN-GRU	0.1094	0.0263	0.1623	0.0346	060.21	0.2439	0.1961	0.4428	0.0509	0.20
Temperature(°C)	LSTM	0.0663	0.0053	0.0728	0.0026	118.71	0.0672	0.0067	0.0818	0.0025	0.36
	GRU	0.3977	0.1607	0.4009	0.0157	120.67	0.4373	0.1943	0.4408	0.0160	0.32
	CNN	0.1138	0.0333	0.1825	0.0046	023.76	0.1441	0.0349	0.1867	0.0052	0.12
	CNN-LSTM	0.0166	0.0006	0.0246	0.0007	043.21	0.0216	0.0009	0.0303	0.0008	0.18
	CNN-GRU	0.8879	0.9603	0.9799	0.0340	062.96	1.3416	1.8252	1.3510	0.0489	0.20

h_p : $L_r = 0.0008$, $E_p = 50$, $W_z = 30$, $B_z = 32$, and Adam optimizer.

0.0063 and MAPE = 0.0005) outperforms all other models in terms of prediction accuracy. The same performance is repeated for DO (MAE = 0.0566 ml/L, MSE = 0.0065 ml/L, RMSE = 0.0804 ml/L, and MAPE = 0.0145 ml/L) and temperature (MAE = 0.0216 °C, MSE = 0.0009 °C, RMSE = 0.0303 °C, and MAPE = 0.0008 °C) prediction as well. Also, the computation time required for CNN-LSTM is around 33% of the computation time needed by LSTM and GRU models. CNN model has lesser computational time when compared to CNN-LSTM. Hence, it can be concluded that CNN-LSTM is the DL model providing better performance for MAC WQ data.

In the evaluation of WQP performance of different DL-RNN models on both the ADAK and MAC water quality datasets the hybrid CNN models outperformed other modes. For the ADAK dataset, the hybrid CNN-LSTM model consistently outperformed all baseline models, achieving lower MAE, MSE, RMSE, and MAPE values. The CNN-LSTM model also demonstrated efficient training times compared to LSTM, making it a superior choice for salinity, pH, DO, and temperature prediction. Similarly, on the MAC dataset, the CNN-LSTM model excelled in prediction accuracy and computation time efficiency, establishing its superiority over LSTM, GRU, CNN, and CNN-GRU for salinity, pH, DO, and temperature forecasting.

5.5 SUMMARY

In this chapter, hybrid DL models, CNN-LSTM and CNN-GRU are proposed for aquaculture WQP. The developed hybrid prediction models were trained and tested on two distinct datasets. The impact of varying the h_p is extensively analysed. For perfor-

mance comparison and further analysis, optimal h_p were used. The performance of these hybrid DL models (CNN-LSTM and CNN-GRU) is compared with baseline DL models (LSTM, GRU, and CNN) in terms of MAE, MSE, RMSE, and MAPE. Results show that the CNN-LSTM hybrid model provides significant improvement in prediction accuracy as well as computation time compared to the baseline DL models, offering a realistic solution for predicting WQ parameters in smart aquaculture.